A PROJECT REPORT

ON

SkillBridge AI: Machine Learning Based Student Placement Prediction and Employability Analysis

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
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Abstract

This project develops a machine-learning classifier for predicting student placement status as placed or not placed using academic performance, employability skills, and engagement indicators. The faculty-provided Placement Prediction dataset contains 50,000 student records and 32 columns. Exploratory data analysis was performed to inspect data quality, target distribution, academic-score patterns, and relationships between student attributes and placement outcomes. A preprocessing pipeline uses median imputation and standard scaling for numerical features, and most-frequent imputation and one-hot encoding for categorical features. A Random Forest classifier was trained on an 80% training split and evaluated on a stratified 20% test split. The model achieved 91.74% accuracy and a ROC-AUC score of 0.9777 on the test set. The results indicate that semester SGPA, mock-interview score, CGPA, and project experience are influential predictors in this dataset. The system is intended as a decision-support aid for career-readiness support, not as a definitive judgement of an individual student.

Keywords: Student placement prediction, Machine learning, Random Forest, Exploratory Data Analysis, Employability, Classification

1. Introduction

1.1 Purpose of the Project

Student placement outcomes depend on several interacting factors, including semester-wise academic performance, cumulative grade point average, coding and aptitude assessments, communication and soft skills, internships, projects, certifications, and interview readiness. Students and placement cells often review these indicators manually. Such review can be time-consuming when the student population is large, and it may be difficult to consistently identify students who could benefit from targeted preparation early enough.

A data-driven prediction system can help summarize patterns in historical student records and provide an additional signal for career-readiness planning. SkillBridge AI frames this as a support tool for students, faculty, and placement coordinators: predictions may guide mentoring, mock interviews, technical practice, or internship planning. The dataset used here was provided for academic project work; model findings describe patterns in this dataset and should not be treated as causal claims or guaranteed future employment outcomes.

1.2 Problem Statement

The task is binary supervised classification. Given a student record containing academic, demographic, skill, and engagement attributes, the model predicts PlacementStatus, where 1 represents placed and 0 represents not placed. The expected output is a predicted class and, where available, a probability estimate that can be used to prioritize supportive interventions. Salary Package, StudentID, and IsAnomaly are excluded from the predictor set to avoid identifier use, post-outcome information, and a dataset-specific anomaly flag.

1.3 Existing Methods and Disadvantages

Traditional placement assessment often relies on manual screening, fixed eligibility cut-offs, or separate reviews of CGPA, backlogs, and test scores. These approaches are easy to explain but may not capture non-linear interactions between academic performance, internships, project work, and interview scores. A single threshold can also overlook students whose strengths are distributed across several indicators.

Machine-learning approaches can model more complex patterns, but they also introduce risks. Models can overfit historical data, learn dataset-specific correlations, or perform unevenly across student groups. High accuracy alone does not establish fairness, causality, or performance on future cohorts. Careful feature selection, leakage prevention, stratified evaluation, explainability, and validation on independent data are therefore important before any real deployment.

1.4 Proposed System

The proposed workflow audits the dataset, explores target and feature distributions, separates the target from the predictors, and removes StudentID, Salary Package, and IsAnomaly from the inputs. Numerical missing values are imputed with the median and scaled using StandardScaler; categorical missing values are imputed with the most frequent value and encoded with OneHotEncoder. These transformations are placed inside a scikit-learn pipeline and fitted on the training data. A Random Forest classifier with 150 trees, maximum depth 15, minimum split size 5, balanced class weights, and random_state 42 is trained and evaluated using accuracy, class-wise precision/recall/F1, ROC-AUC, and a confusion matrix.

1.5 Research Objectives(atleast 4)

1. Prepare a suitable dataset for the identified ML problem.
2. Perform systematic preprocessing and exploratory analysis.
3. Engineer/select relevant features while preventing data leakage.
4. Develop and train suitable machine learning models.
5. Evaluate and compare models using appropriate metrics.
6. Identify the best-performing model and discuss its practical applicability.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

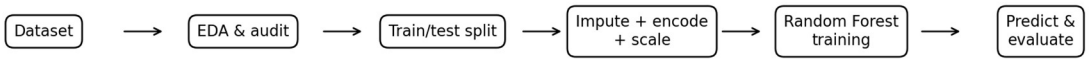
S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Prognostication of the placement of students applying machine learning algorithms	Basil (2023), BOHR International Journal of Internet of Things, Artificial Intelligence and Machine Learning, 2(1), 26–30. DOI: 10.54646/bijiam.2023.14	Placement eligibility is difficult to assess consistently from student records.	Predict placement using academic and extracurricular indicators.	Compares ML classification approaches including Logistic Regression and K-NN.	Reports that ML can support placement prediction from student databases.	Test on larger and more diverse institutional datasets.
2	OPT-BAG Model for Predicting Student Employability	Vo, Nguyen, & Le (2023), Computers, Materials & Continua, 76(2), 1555–1568. DOI: 10.32604/cmc.2023.039334	Single classifiers may not capture employability patterns robustly.	Develop an optimized bagging model for employability prediction.	Optimization of bagging classifiers (OPT-BAG).	Ensemble learning is proposed to improve employability prediction.	Validate across institutions and compare computational cost.
3	Predicting Student Placements Using Machine Learning Algorithms	Hassan & Goyal (2023), Journal of Propulsion Technology, 44(4). DOI: 10.52783/tjjpt.v44.i4.1098	Manual assessment does not scale well across student profiles.	Compare models for placed/not-placed classification.	Decision Tree, Random Forest, XGBoost, and KNN with EDA.	The authors report Random Forest as the strongest model in their study.	Report independent validation and detailed reproducibility.
4	Predicting College Students' Placements Based on Academic Performance Using Machine Learning Approaches	(2023), International Journal of Modern Education and Computer Science, 15(6). DOI: 10.5815/ijmecs.2023.06.01	Academic and activity variables interact in placement outcomes.	Predict placement from academic markers and student activities.	Logistic Regression, Gaussian Naive Bayes, Random Forest, SVM, KNN.	Compares several standard classifiers for placement prediction.	Assess generalization across colleges and cohorts.
5	Classification Techniques Using Machine Learning for Graduate Student Employability Predictions	Haque et al. (2024), International Journal on Advanced Science, Engineering and Information Technology, 14(1), 45–56. DOI: 10.18517/ijaseit.14.1.19549	Graduate employability is a complex outcome influenced by multiple factors.	Evaluate classification techniques for employability prediction.	Comparative machine-learning classification study.	Supports using data-driven analysis to understand graduate employability.	Include fairness, calibration, and temporal validation.
6	Predictive Analysis of Campus Placement of Student Using Machine Learning Algorithms	Agrawal & Kadam (2024), Journal of IoT and Machine Learning. DOI: 10.48001/joitml.2024.1213-18	Placement teams need reliable forecasts from student academic data.	Forecast campus placement using academic factors.	Compares ML approaches including tree-based and linear models.	Highlights the relevance of algorithm comparison in placement prediction.	Expand datasets and validate outside the source institution.

7	Predicting Post-Internship Employability Using Ensemble Machine Learning Approach	Kahlik & Al-Hababi (2024), Journal of Cognitive Sciences and Human Development, 10(2), 87–101. DOI: 10.33736/jcshd.7518.2024	Academic results alone may not capture work readiness after internship.	Study internship-related factors in employability outcomes.	Compares machine-learning algorithms including ensemble methods.	Internship duration, training, and work experience merit consideration.	Collect more internship-quality measures and external cohorts.
8	Fair and Transparent Student Placement Predictions: A Machine Learning and Explainable AI (XAI) Approach	Jindia & Chawla (2024), ShodhKosh: Journal of Visual and Performing Arts. DOI: 10.29121/shodhkosh.v5.i4.2024.5996	Opaque predictions can be difficult to trust and may reproduce bias.	Improve transparency and fairness of career-path prediction.	Machine learning with explainable AI methods such as LIME/SHAP.	Interpretability and bias mitigation are important alongside accuracy.	Audit subgroup fairness and explanation stability.
9	Developing a Predictive Model for Engineering Graduates Placement Using a Data-Driven Machine Learning Approach	Molla et al. (2024), institutional research record, Middle East Technical University repository. DOI details not present in the supplied bibliographic record.	Engineering placement requires multi-factor, data-driven modeling.	Predict engineering graduates' placement outcomes and related job preferences.	Multiple ML algorithms; customized Random Forest and PCA variants reported.	The repository summary reports 89% prediction rate for customized RF.	Verify the final published bibliographic record and validate externally.
10	AI-Driven Employability Prediction: Integrating Machine Learning and Educational Analytics for Student Placements	Bhujade, Khy, & Dholay (2026), International Journal of Research in Engineering, Science and Management, 9(1). DOI: 10.65138/ijresm.v9i1.3404	Students and institutions need actionable links between skills and employment.	Predict employability and potential job-role/company fit.	SVM, Random Forest, Gradient Boosting, and XGBoost.	Combines academic, technical, internship, and leadership attributes.	Evaluate calibration, fairness, and deployment on real cohorts.

3. Methodology

3.1 Proposed System Flow Diagram

The implemented workflow is summarized below. Hyperparameter search was not performed in the supplied notebook; the Random Forest settings were specified directly.



Recommended flow: Dataset → Data Cleaning → Missing-Value Treatment → Encoding → Feature Engineering/Selection → Train/Test Split → Model Training → Hyperparameter Tuning → Prediction → Evaluation → Final Model.

3.2 Dataset Description

The dataset supplied for this academic project is titled "Placement Prediction – 50,000 records" and is stored as a CSV file. It contains 50,000 rows and 32 columns, including student demographics, city, engineering stream, specialization, semester-wise SGPA, CGPA, attendance, internships, projects, workshops, certifications, publications, aptitude/coding/mock-interview scores, extracurricular participation, CGPA tier, placement status, an anomaly flag, and salary package. The target is PlacementStatus (binary: 0 = not placed, 1 = placed). The dataset has 32,856 placed records (65.71%) and 17,144 not-placed records (34.29%). It is treated as a faculty-provided academic dataset; no external collection process or public provenance was specified in the supplied files.

3.3 Experimental Setup

The supplied notebook is written for Google Colab and uses Python with pandas, NumPy, Matplotlib, Seaborn, and scikit-learn. The notebook imports files from the Colab upload interface and uses a CSV path under /content. The exact Python and package versions and the hardware/runtime specification are not recorded in the notebook, so they are not asserted here. The workflow can also be run in a compatible Jupyter environment after updating the dataset path.

3.4 Data Preprocessing

- Data inspection and schema validation: confirmed 50,000 records and 32 columns with mixed numeric and categorical data.
- Duplicate removal: 0 fully duplicated rows were detected; no rows required removal.
- Missing-value analysis and imputation: missing values occur in five columns—MockInterviewScore (4,957), Workshops (4,488), AptitudeTestScore (4,029), SoftSkillsRating (3,526), and CodingTestScore (2,976), totalling 19,976 missing cells. The training pipeline applies median imputation to numeric features and most-frequent imputation to categorical features.
- Outlier analysis/treatment: EDA includes distributions and box plots for CGPA, attendance, and coding scores. The notebook does not apply a separate outlier-removal rule; observations were not deleted solely on the basis of extreme values.
- Categorical encoding: OneHotEncoder(handle_unknown="ignore") transforms the eight categorical predictor columns.
- Numerical scaling/standardization: StandardScaler is applied to the 20 numeric predictor columns after median imputation.
- Target-label preparation: PlacementStatus is separated as y; predictor matrix X excludes PlacementStatus, Salary Package, StudentID, and IsAnomaly, leaving 28 input columns before encoding.
- Train/test split and leakage prevention: an 80/20 split with random_state=42 and stratify=y yields 40,000 training rows and 10,000 test rows. The preprocessing transformer is fitted as part of the pipeline on training data. Salary Package and the identifier/anomaly fields are excluded.

3.5 Exploratory Data Analysis

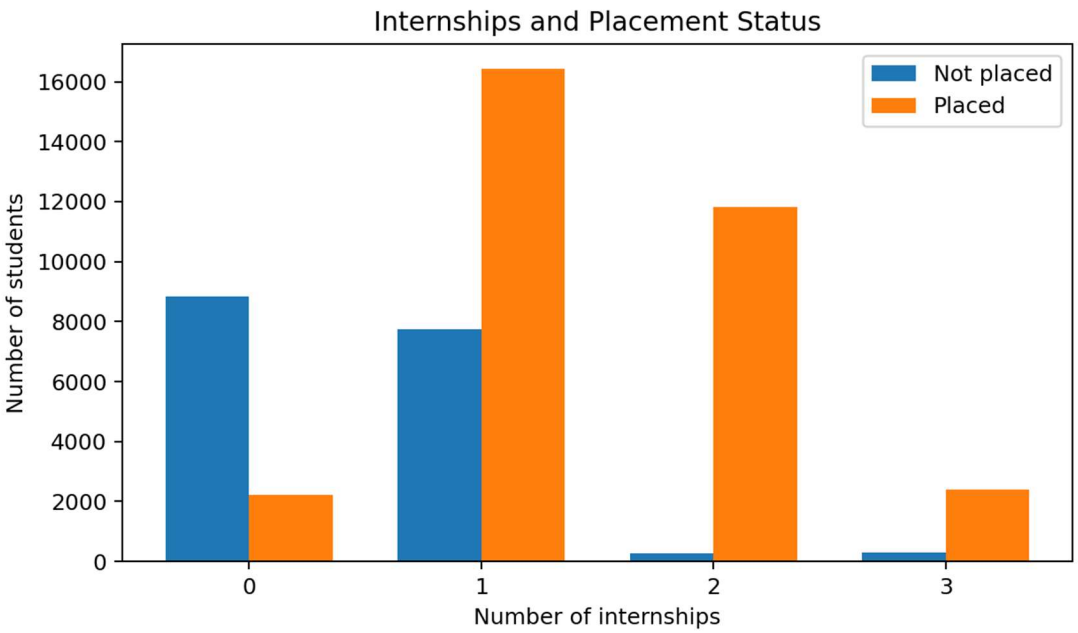
EDA in the supplied notebook includes placement-status distribution, gender and engineering-stream distributions, CGPA histogram, attendance/placement and CGPA/placement box plots, coding-score/placement box plot, internship count by placement status, numeric correlation heatmap, and correlations with PlacementStatus. The target is moderately imbalanced: 65.71% placed and 34.29% not placed. Five predictor columns contain missing values and there are no duplicate rows. The strongest raw numeric correlations with PlacementStatus include SGPA_Sem8 (r = 0.707), SGPA_Sem7 (r = 0.702), SGPA_Sem6 (r = 0.696), and MockInterviewScore (r = 0.668). Salary Package also has a high correlation (r = 0.774), but it is excluded because it can reflect information available only after placement and would create leakage.

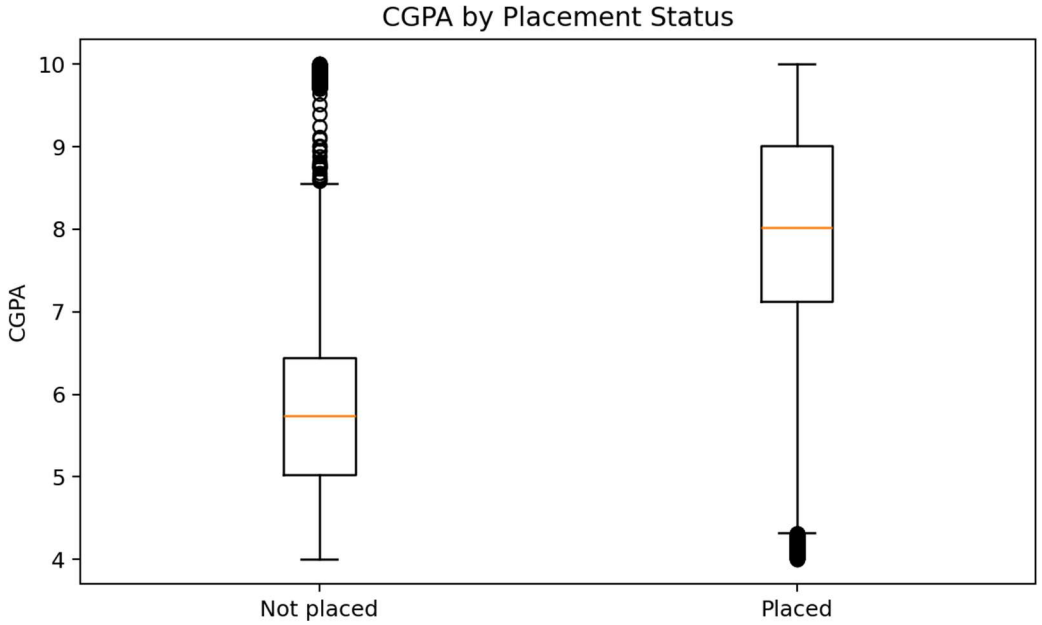
Figure 2. Distribution of the target variable PlacementStatus.

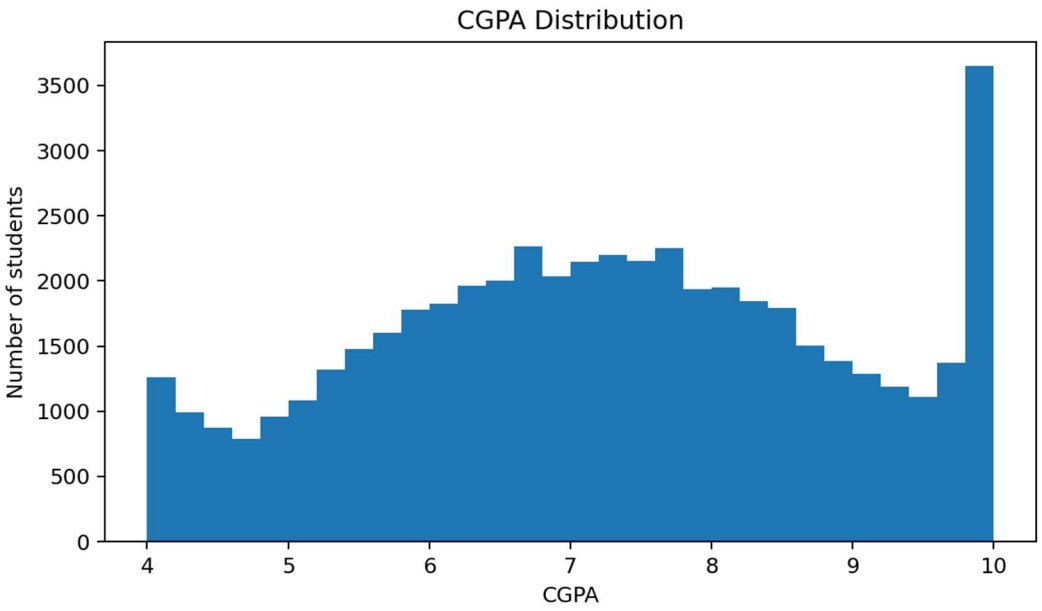
Figure 3. CGPA distribution across the dataset.

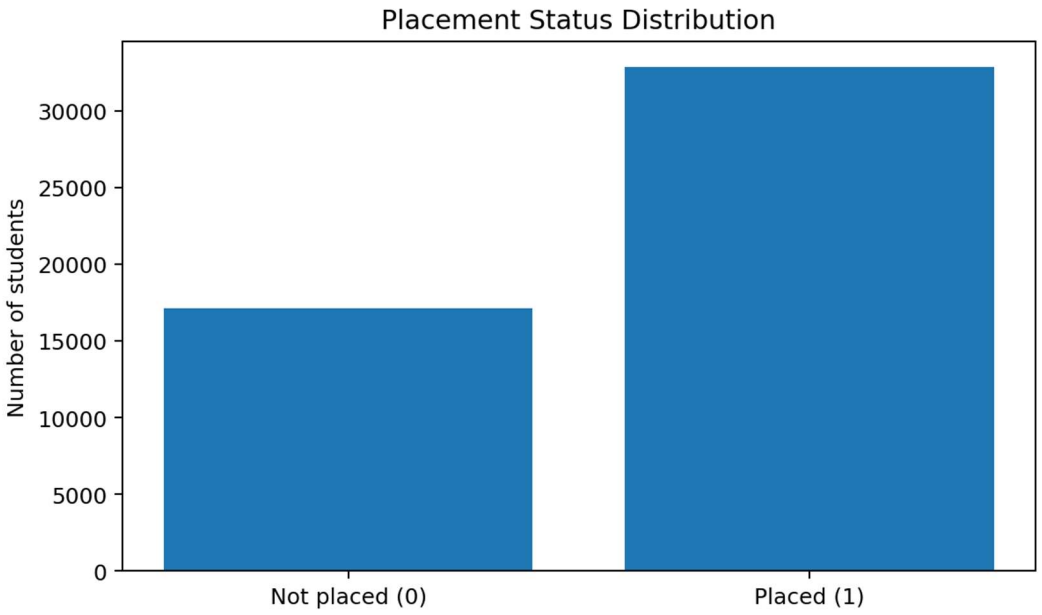
Figure 4. CGPA by placement status.

Figure 5. Internship counts grouped by placement status.


Internships and Placement Status
Not placed
Placed
Number of students
16000
14000
12000
10000
8000
6000
4000
2000
0
0
1
2
3
Number of internships


CGPA by Placement Status
CGPA
10
9
8
7
6
5
4
Not placed
Placed





3.6 Feature Engineering and Feature Selection (if applicable)

The supplied notebook does not create new domain-derived columns. It uses the existing semester-wise SGPA, CGPA, CGPA_Tier, attendance, internships, projects, workshops, certifications, publications, aptitude, soft-skills, coding, mock-interview, and extracurricular attributes. One-hot encoding expands categorical variables into indicator columns, while numerical scaling standardizes numeric predictors. StudentID is excluded as an identifier; Salary Package is excluded as likely post-outcome information; IsAnomaly is excluded as a dataset-specific flag that should not serve as a student characteristic.

3.7 Model Building and Training

The notebook trains one Random Forest classifier using scikit-learn. The configured parameters are n_estimators=150, max_depth=15, min_samples_split=5, class_weight="balanced", random_state=42, and n_jobs=-1. Random Forest aggregates decision trees and can model non-linear relationships among mixed student attributes. The preprocessing and

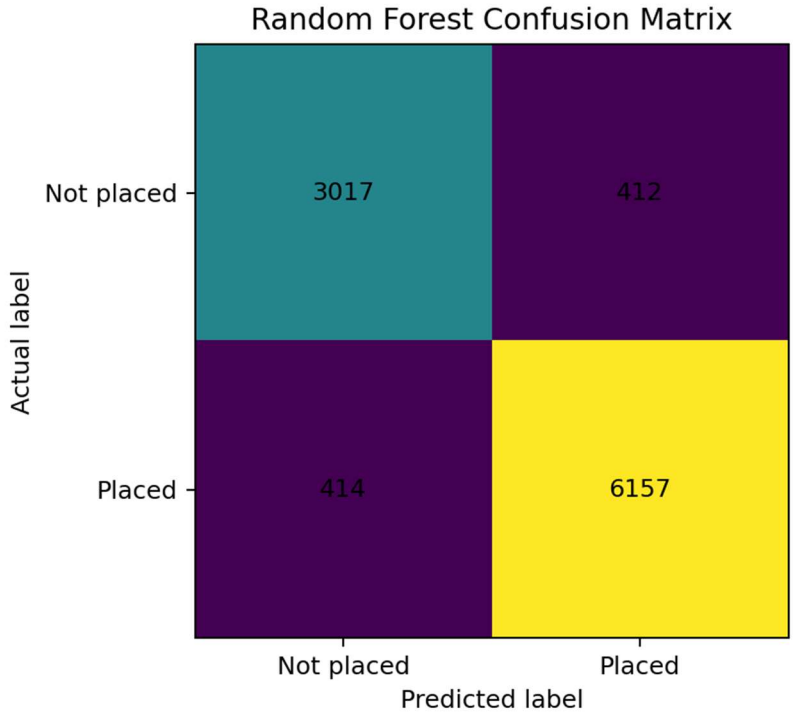
classifier are combined in one Pipeline, and the model is fitted using the 40,000-row training set. The supplied notebook does not train a separate baseline or additional classifier.

3.8 Hyperparameter Tuning(if applicable)

No GridSearchCV, RandomizedSearchCV, or cross-validation-based hyperparameter search is present in the supplied notebook. The reported metrics therefore correspond to the explicitly configured Random Forest parameters and the fixed stratified hold-out test set. A future iteration should tune parameters using cross-validation confined to the training partition.

3.9 Model Evaluation

Evaluation uses accuracy, class-wise precision, recall and F1-score, ROC-AUC from predicted probabilities, and a confusion matrix. On the 10,000-row test set, accuracy is 0.9174 (91.74%) and ROC-AUC is 0.97766. For class 1 (placed), precision, recall, and F1-score are each approximately 0.94. The confusion matrix, with rows as actual and columns as predicted, is [[3017, 412], [414, 6157]].



4. Results and Discussion

4.1 Preprocessing Results

The raw dataset has 50,000 rows and 32 columns and contains 19,976 missing cells across five columns. The notebook detects no duplicated rows. The predictor matrix contains 28 columns before preprocessing. Median imputation and standard scaling are applied to numeric columns, while most-frequent imputation and one-hot encoding are applied to categorical columns. With the categorical levels in this dataset, the transformed feature count is generated by the fitted encoder at training time; the notebook does not print a final transformed-column count or a before/after missing-value table, so a precise encoded count is not claimed in this report.

4.2 Model Performance

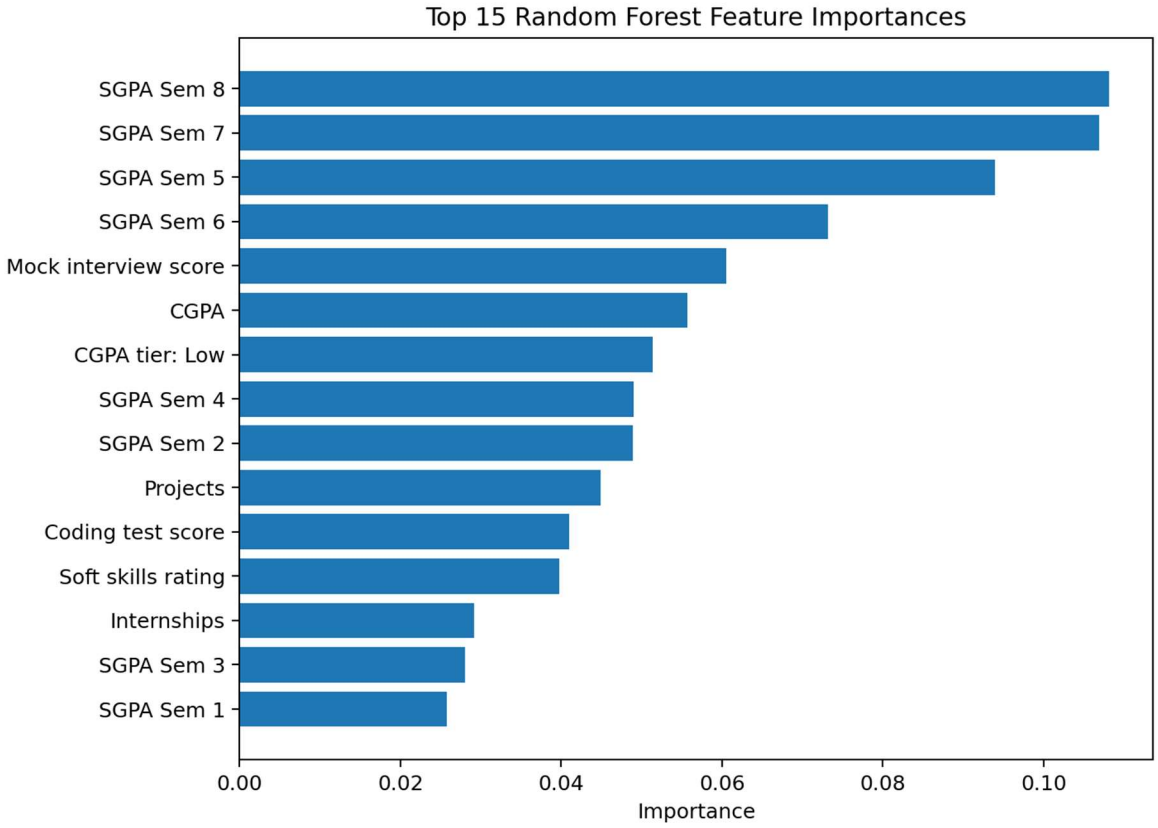
Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Random Forest (class 1: placed)	91.74%	0.94	0.94	0.94	0.9777
Class 0 (not placed)	—	0.88	0.88	0.88	—

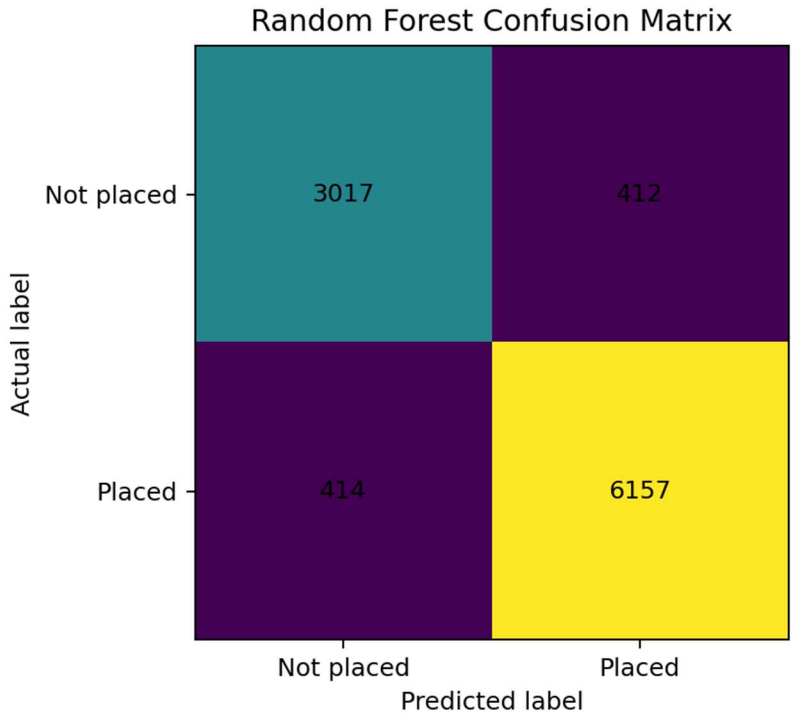
Macro average	—	0.91	0.91	0.91	—
Weighted average	—	0.92	0.92	0.92	—
Other models	Not evaluated	—	—	—	—

The confusion matrix and feature-importance chart below are taken from the supplied notebook outputs. The confusion matrix shows 3,017 true negatives, 412 false positives, 414 false negatives, and 6,157 true positives. Thus, the model correctly classifies 9,174 of 10,000 test records. Feature importance is highest for SGPA_Sem8 (0.1082), SGPA_Sem7 (0.1069), SGPA_Sem5 (0.0940), and SGPA_Sem6 (0.0732), followed by MockInterviewScore (0.0605) and CGPA (0.0557). These are model-specific importance scores, not proof that changing one feature causes placement.

Figure 6. Confusion matrix for the Random Forest classifier.

Figure 7. Top 15 features ranked by Random Forest impurity-based importance.





[Insert relevant training/validation curves, confusion matrix, ROC curve, feature importance, prediction-vs-actual plot, residual plot, etc.]

The model obtains 91.74% test accuracy, with similar false-positive and false-negative counts (412 and 414). The class-1 precision/recall/F1 are each about 0.94, while the class-0 precision/recall/F1 are each about 0.88. ROC-AUC of 0.9777 indicates strong ranking discrimination on this test split. The top feature importances are dominated by later-semester academic scores and interview readiness. However, only one algorithm was evaluated in the provided notebook, so the result cannot establish that Random Forest is superior to other models. Results are also tied to this dataset and split; external validation, subgroup fairness checks, probability calibration, and monitoring would be required before operational use.

5. Conclusion

This project applied a Random Forest classifier to the binary student-placement prediction task using a faculty-provided dataset of 50,000 records and 32 columns. EDA identified target imbalance, missing values in five skill/engagement fields, and meaningful associations between placement and semester grades, interview scores, coding skills, projects, and internships. A leakage-conscious feature set excluded StudentID, Salary Package, IsAnomaly, and the target. A pipeline handled missing values, encoding, and scaling, and a stratified 80/20 split was used for evaluation. The classifier achieved 91.74% accuracy and 0.9777 ROC-AUC on 10,000 test examples. These results support further exploration of ML-assisted career-readiness tools, while the dataset-specific nature of the findings and the absence of external validation must be acknowledged.

6. Future Work

- Use larger and more diverse cohorts from multiple institutions and graduating years, with documented collection methods and consent.
- Compare Logistic Regression, Decision Tree, Gradient Boosting, XGBoost, and calibrated ensemble models on the same stratified split and through cross-validation.
- Perform cross-validated hyperparameter optimization and report confidence intervals and repeated-split performance.
- Use SHAP or other explainability methods and test whether explanations are stable for individual students.

- Evaluate subgroup performance, fairness, calibration, robustness to missing/noisy inputs, and drift across new graduating cohorts.
- Integrate the validated model into SkillBridge AI as a decision-support feature with privacy safeguards and clear user-facing limitations.
- Validate the approach on a held-out dataset from a different institution before making real-world claims.

7. References

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[5] [Authors], "[Paper title]," [Journal/Conference], vol. [x], no. [x], pp. [xx–xx], [year], doi: [DOI].

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[8] [Authors], "[Paper title]," [Journal/Conference], vol. [x], no. [x], pp. [xx–xx], [year], doi: [DOI].

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[10] [Authors], "[Paper title]," [Journal/Conference], vol. [x], no. [x], pp. [xx–xx], [year], doi: [DOI].

Signature of the Guide Course Instructor